

Summary and Recommendation

Project Overview

This project analyzes the **Brazilian E-Commerce (Olist) dataset** using **SQL and Python** to uncover customer purchasing behavior, sales performance, revenue trends, and seller contributions. The objective was to transform raw transactional data into meaningful business insights that can support data-driven decision-making for an e-commerce platform.

Objectives

- Analyze customer distribution across different cities and states.
 - Measure order volume and sales performance over time.
 - Identify top-performing product categories and sellers.
 - Understand customer purchasing patterns and payment behavior.
 - Evaluate revenue contribution and business growth trends.
 - Apply advanced SQL concepts such as CTEs, Window Functions, Ranking, and Aggregations.
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Data Understanding and Preparation

The analysis was performed on multiple interconnected Olist datasets containing:

- Customer information
- Orders data
- Products data
- Order items
- Payments data
- Seller information
- Geolocation data

Data Preparation Steps

- Connected MySQL database with Python.
- Imported and explored all tables.
- Verified data consistency across related tables.

- Performed joins between customers, orders, products, sellers, and payments datasets.
 - Aggregated transactional data for business analysis.
 - Created derived metrics such as revenue, sales percentage, rankings, moving averages, and growth rates.
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Exploratory Data Analysis (EDA)

Customer Location Analysis

The analysis identified all unique customer cities and counted customers across different states.

Findings

- Customers were distributed across numerous cities and states.
- Certain states contributed significantly larger customer bases than others.

Business Insight

Regions with higher customer concentrations can be prioritized for marketing campaigns and logistics optimization.

Orders Analysis

Order volume was analyzed over time.

Findings

- A substantial number of orders were placed during 2017 and 2018.
- Monthly order trends showed fluctuations throughout the year.

Business Insight

Understanding seasonal order patterns helps improve inventory planning and resource allocation.

Sales by Product Category

Total sales generated by each product category were calculated.

Findings

- Some product categories generated significantly higher revenue than others.
- Revenue distribution was concentrated among a limited number of categories.

Business Insight

High-performing categories should receive greater promotional focus and inventory investment.

Payment Behavior Analysis

The percentage of orders paid through installment payments was examined.

Findings

- A large proportion of customers preferred installment-based payments.
- Flexible payment options played an important role in customer purchasing decisions.

Business Insight

Maintaining installment payment options can improve conversion rates and customer satisfaction.

Customer City Purchase Analysis

The average number of products purchased per order was analyzed by customer city.

Findings

- Purchasing behavior varied across cities.
- Some cities demonstrated higher average order sizes than others.

Business Insight

Location-specific marketing strategies can be developed based on purchasing behavior.

Revenue Contribution Analysis

The percentage contribution of each product category to total revenue was calculated.

Findings

- A small number of categories contributed a major share of total revenue.

- Several categories generated relatively low revenue despite having sales volume.

Business Insight

Businesses should focus on high-revenue categories while evaluating underperforming categories for improvement opportunities.

Product Price vs Purchase Frequency

The relationship between product pricing and purchase frequency was explored.

Findings

- Certain lower-priced products experienced higher purchase frequencies.
- Premium-priced products generated revenue with lower purchase volumes.

Business Insight

A balanced product portfolio containing both high-volume and high-margin products can maximize profitability.

Seller Performance Analysis

Revenue generated by each seller was calculated and sellers were ranked accordingly.

Findings

- Revenue distribution among sellers was uneven.
- A small group of sellers contributed a significant portion of total sales.

Business Insight

Top-performing sellers can be rewarded and supported to sustain business growth.

Customer Order Value Trends

A moving average of customer order values was calculated to study purchasing patterns.

Findings

- Customer spending behavior changed over time.

- Moving averages helped smooth short-term fluctuations and reveal long-term trends.

Business Insight

Tracking spending trends enables personalized recommendations and customer retention initiatives.

Cumulative Sales Analysis

Cumulative monthly sales were analyzed using window functions.

Findings

- Sales consistently increased over time.
- The cumulative trend highlighted overall business growth.

Business Insight

Cumulative sales metrics provide a clear picture of long-term business performance.

Year-over-Year Growth Analysis

Annual sales growth rates were calculated.

Findings

- Revenue growth varied between years.
- Certain years experienced stronger sales performance than others.

Business Insight

Growth analysis helps management evaluate business expansion and strategic effectiveness.

Key Business Insights

The analysis revealed several important insights:

- Customer demand is concentrated in specific regions.
- Installment payments are highly popular among customers.
- A few product categories drive the majority of revenue.

- Seller performance is uneven, with top sellers contributing significantly more revenue.
 - Customer purchasing behavior differs across cities.
 - Sales exhibit seasonal patterns throughout the year.
 - Overall revenue demonstrates positive long-term growth.
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Business Recommendations

Recommendation 1: Focus on High-Revenue Categories

Allocate marketing and inventory resources toward categories contributing the largest share of revenue.

Recommendation 2: Strengthen Seller Relationships

Provide incentives and support programs for top-performing sellers.

Recommendation 3: Optimize Regional Marketing

Target states and cities with strong customer bases using location-specific campaigns.

Recommendation 4: Maintain Flexible Payment Options

Continue offering installment payment methods to improve customer conversion rates.

Recommendation 5: Monitor Growth Trends

Regularly track cumulative sales and year-over-year growth to identify opportunities and risks.

Recommendation 6: Improve Product Portfolio Strategy

Balance high-volume products with high-margin premium products to maximize profitability.



Conclusion

This E-Commerce Sales Analysis successfully leveraged SQL and Python to explore customer behavior, sales trends, revenue distribution, payment preferences, and seller performance. The results demonstrate that revenue growth is driven by a combination of high-performing product categories, strong seller contributions, and widespread customer adoption of installment payments. By focusing on regional opportunities, supporting top sellers, and optimizing product

and pricing strategies, the business can improve profitability, customer satisfaction, and long-term growth.